Assignment 3: AI System Implementation

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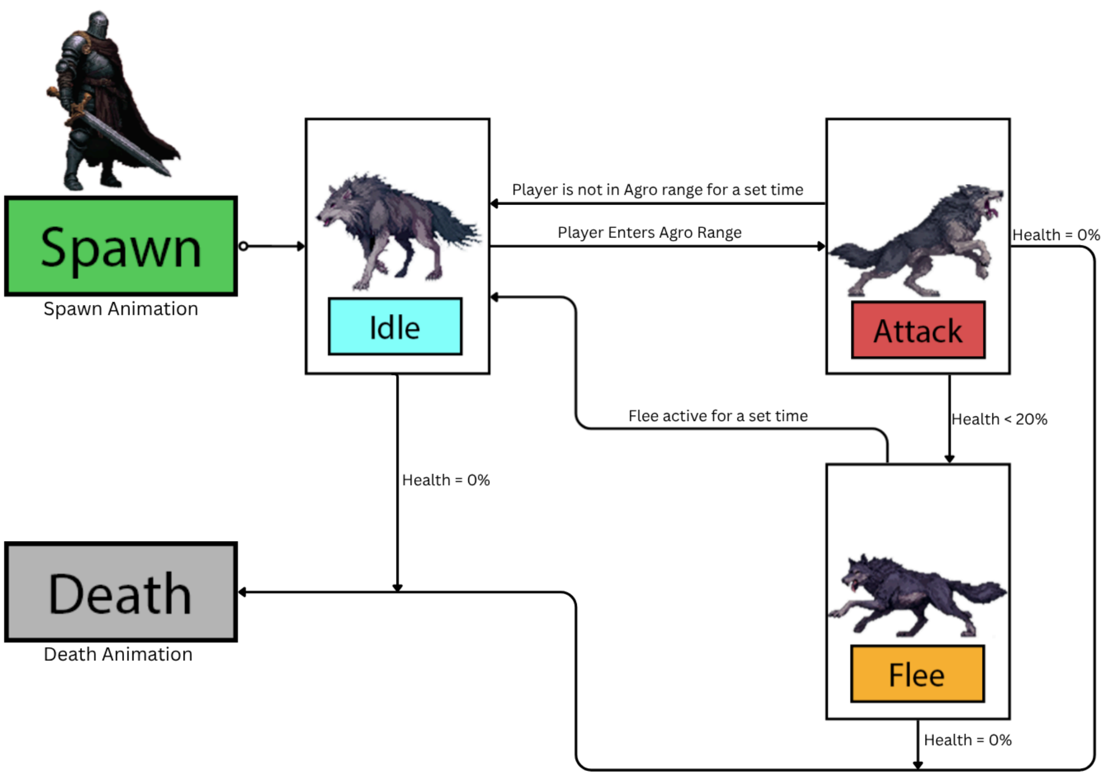
Game Systems Development - GAME-10020-01

Introduction:

The concept of this design is to create a pack of wolves that exercise some pack mentality. Wolves flee when low, unless they are surrounded by at least 3 other wolves.

Machine Diagram:



Transition Details:

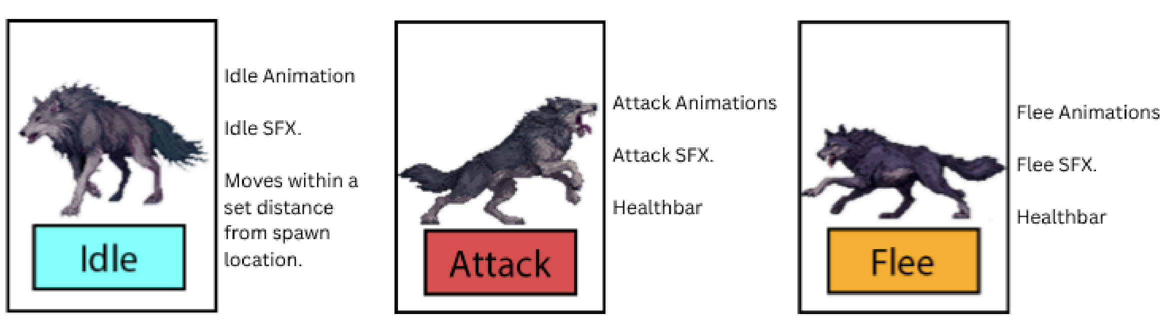
Idle - Attack: Player is within view of the wolf's cone shaped trigger view area.

Attack - Flee: If any wolf falls below 20% hp, they will enter the Flee state unless 3 other wolves, or more, are nearby.

Flee - Idle: After a set amount of time, wolf health will restore, re-entering Idle.

Any state - Death: If the wolf's hp falls to 0%, they die.

States and Behaviors:



Example Scenario:

Development Challenges:

Throughout development of this assignment, my scope was constantly changing. My initial design ended up feeling too simple, then as I went to implement what I had pivoted to, I realised it actually could become more complex.

My fundamental problem when developing is that I am designer brained, and want to strive for a complete finished product, prioritizing things I shouldn't

This happened constantly with this assignment, and the constant pivots made it impossible to update this design document accurately. There are additional mechanics within the session that are not included here, and there was no time to clearly explore them.

I incorporated a **howl**, that will trigger on a random chance, when and wolf takes damage. If any other wolf hears that sound, it will trigger a state not mentioned called 'Hunt' which will essentially seek out the sounds area, with a generous threshold, **rather** than flee. The audio trigger seemed more interesting.

Naming conventions are also inaccurate in this document, having used mostly the naming conventions from class, rather than this document, so as to not need to replace large amounts of code. A very difficult assignment, but I learned a great deal.